THE QUOTIENT FUNCTOR SATISFIES THE RIM-SCHLESSINGER CONDITION ON PUSHOUTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition situation-q (canonical tag 082M). cited at statement line 2215. Standing situation.
- Lemma more-morphisms-lemma-pushout-along-thickening (canonical tag 07RT). cited at proof line 2224. The pushout square.
- Lemma spaces-pushouts-lemma-space-over-pushout-flat-modules (canonical tag 08KV). cited at proof line 2241. Construction of A'.
- Lemma more-algebra-lemma-module-over-fibre-product-bis (canonical tag 08IG). cited at proof line 2262. First arrow surjective.
- Lemma more-algebra-lemma-surjection-module-over-fibre-product (canonical tag 08KJ). cited at proof lines 2263-2265. Second arrow surjective.
- Remark more-algebra-remark-relative-modules-over-fibre-product (canonical tag 08KR). cited at proof line 2272. Finitely presented case.

Source reference. Original definition 082M: sources/stacks-chow/tags.txt, source lines 10257–10257. Cited at statement line 2215: the lemma is phrased in this situation.

Source reference. Original lemma 07RT: sources/stacks-chow/tags.txt, source lines 9878–9878. Cited at proof line 2224: the pushout square exists.

Source reference. Original lemma 08KV: sources/stacks-chow/tags.txt, source lines 10898–10898. Cited at proof line 2241: produces A' flat over Y'.

Source reference. Original lemma 08IG: sources/stacks-chow/tags.txt, source lines 10814–10814. Cited at proof line 2262: first arrow surjective.

Source reference. Original lemma 08KJ: sources/stacks-chow/tags.txt, source lines 10887–10887. Cited at proof lines 2263-2265: second arrow surjective.

Source reference. Original remark 08KR: sources/stacks-chow/tags.txt, source lines 10894–10894. Cited at proof line 2272: the Q^fp case in the algebra setting.

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Lemma 0.2. *In Situation 082M. Let*

$$\begin{array}{ccc} Z & \longrightarrow & Z' \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y' \end{array}$$

be a pushout in the category of schemes over B where $Z \to Z'$ is a thickening and $Z \to Y$ is affine, see More on Morphisms, Lemma 07RT. Then the natural map

$$Q_{\mathcal{F}/X/B}(Y') \longrightarrow Q_{\mathcal{F}/X/B}(Y) \times_{Q_{\mathcal{F}/X/B}(Z)} Q_{\mathcal{F}/X/B}(Z')$$

is bijective. If $X \to B$ is locally of finite presentation, then the same thing is true for $Q^{fp}_{\mathcal{F}/X/B}$.

Proof. Let us construct an inverse map. Namely, suppose we have $\mathcal{F}_Y \to \mathcal{A}$, $\mathcal{F}_{Z'} \to \mathcal{B}'$, and an isomorphism $\mathcal{A}|_{X_Z} \to \mathcal{B}'|_{X_Z}$ compatible with the given surjections. Then we apply Pushouts of Spaces, Lemma 08KV to get a quasi-coherent module $\mathcal{A}'$ on $X_{Y'}$ flat over Y'. Since this sheaf is constructed as a fibre product (see proof of cited lemma) there is a canonical map $\mathcal{F}_{Y'} \to \mathcal{A}'$. That this map is surjective can be seen because it factors as

$$\begin{array}{c} \mathcal{F}_{Y'} \\ \downarrow \\ (X_Y \to X_{Y'})_*\mathcal{F}_Y \times_{(X_Z \to X_{Y'})_*\mathcal{F}_Z} (X_{Z'} \to X_{Y'})_*\mathcal{F}_{Z'} \\ \downarrow \\ \mathcal{A}' = (X_Y \to X_{Y'})_*\mathcal{A} \times_{(X_Z \to X_{Y'})_*\mathcal{A}|_{X_Z}} (X_{Z'} \to X_{Y'})_*\mathcal{B}' \end{array}$$

and the first arrow is surjective by More on Algebra, Lemma 08IG and the second by More on Algebra, Lemma 08KJ.

In the case of $Q^{fp}_{\mathcal{F}/X/B}$ all we have to show is that the construction above produces a finitely presented module. This is explained in More on Algebra, Remark 08KR in the commutative algebra setting. The current case of modules over algebraic spaces follows from this by étale localization. □